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# Stability of Psychological Network Models

ECR Retreat DYNAMIC 2026

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## Björn Siepe

Postdoc at Philipps University and Goethe University (soon)

- Time series modeling (incl. dynamic networks), open software, metascience
- Dynamic networks: particularly Bayesian estimation



## Kai Jannik Nehler

Postdoc at Goethe University and Justus Liebig University

- (Cross-sectional) Network estimation under challenging data conditions
- Specifically Ordinal and missing values

# Outline

1 Terminology and Motivation

2 Cross-sectional Networks

3 Longitudinal networks

4 Conclusion

# Two perspectives on network stability




Schematic network.

**Definition**

**Core question**

**Related concepts**

**(Borsboom et al., 2018; Epskamp et al., 2018; Haslbeck et al., 2021a)**

# Two perspectives on network stability




Schematic network.

| | **Stability of estimation** |
|---|---|
| **Definition** | Sensitivity of estimated network features to sampling variation. |
| **Core question** | Would another sample from the same population support similar conclusions? |
| **Related concepts** | precision, robustness |

**(Borsboom et al., 2018; Epskamp et al., 2018; Haslbeck et al., 2021a)**

# Two perspectives on network stability



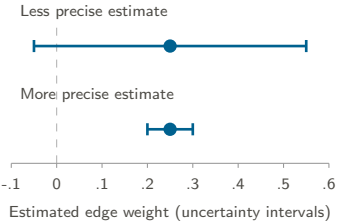

Schematic network.

| | **Stability of estimation** | **Stability across time and groups** |
|---|---|---|
| **Definition** | Sensitivity of estimated network features to sampling variation. | Similarity of underlying network parameters across occasions or populations. |
| **Core question** | Would another sample from the same population support similar conclusions? | Do the same parameters hold at another time or in another population? |
| **Related concepts** | precision, robustness | invariance, generalizability |

**(Borsboom et al., 2018; Epskamp et al., 2018; Haslbeck et al., 2021a)**

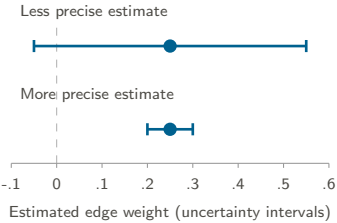
# Why does stability of estimation matter?

- Network connections are estimated rather than directly observed **(Epskamp et al., 2018)**



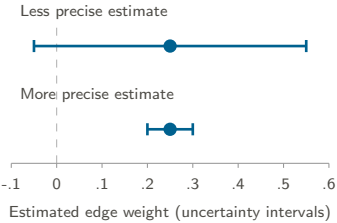
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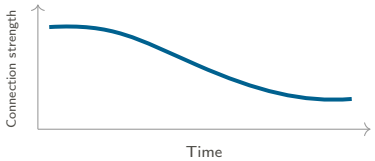



Interpret substantive conclusions in light of estimation precision.

# Why is stability across time and groups interesting?

**Across time** (Haslbeck et al., 2021a)

- Interventions may aim to change relationships between symptoms
- Development and life events may strengthen or weaken these relationships

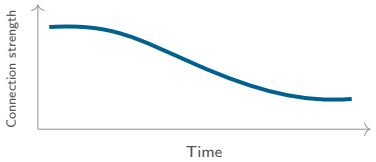



Does the insomnia–fatigue connection change over time?

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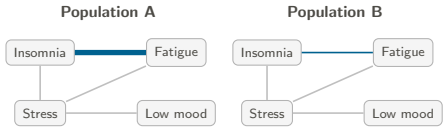
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- Group comparisons help distinguish shared from population-specific relationships

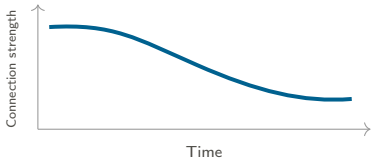



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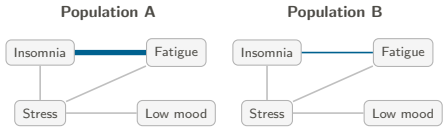
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Is the insomnia–fatigue connection similar across populations?

Stability and change are often themselves be the substantive research question.

# Different estimated networks: evidence of different underlying relationships?




**(Borsboom et al., 2018; Epskamp et al., 2018; Haslbeck et al., 2021a)**

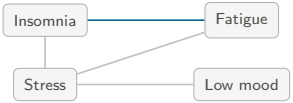
# Different estimated networks: evidence of different underlying relationships?

**Group / Time A**




?

**Group / Time B**




**Substantive perspective**

- Network differences may be meaningful.

**Statistical perspective**

- Estimated networks may differ even when underlying relationships are identical.

**(Borsboom et al., 2018; Epskamp et al., 2018; Haslbeck et al., 2021a)**

# Different estimated networks: evidence of different underlying relationships?




**Substantive perspective**

- Network differences may be meaningful.

**Statistical perspective**

- Estimated networks may differ even when underlying relationships are identical.

## Implications for this talk

- Investigating substantive stability and change
- Quantifying and communicating statistical uncertainty

**(Borsboom et al., 2018; Epskamp et al., 2018; Haslbeck et al., 2021a)**

# Outline

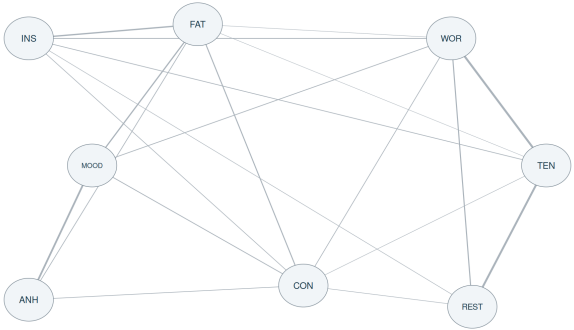
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# From estimated networks to substantive conclusions

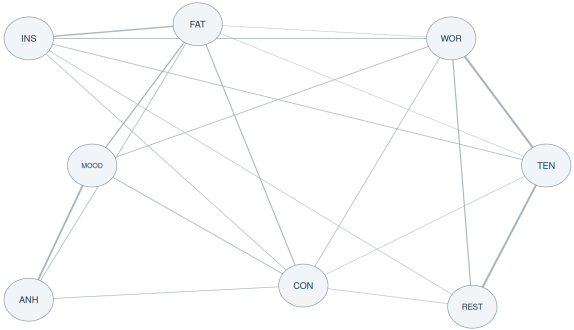



Simulated data; regularized partial correlations.

## Gaussian graphical models **(Epskamp & Fried, 2018)**

- Nodes represent variables; edges represent unique relations
- different claims about associations require different kinds of statistical support
- always including uncertainty due to sampling variability

# From estimated networks to substantive conclusions




Simulated data; regularized partial correlations.

## Gaussian graphical models **(Epskamp & Fried, 2018)**

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## What do we want to learn from this network?

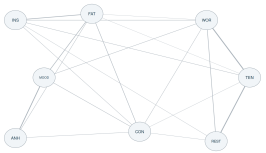
- Which conditional associations characterize the symptom pattern?
- How large are these associations?
- Which associations are supported as present or absent?

# The familiar workflow: What can we conclude?

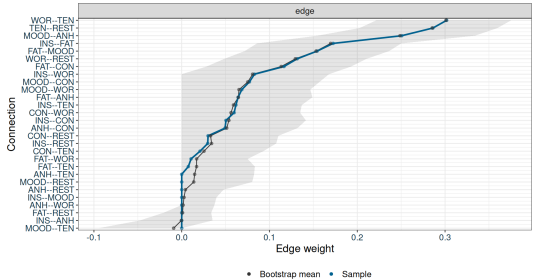
Which associations characterize the pattern? **(Epskamp & Fried, 2018; Epskamp et al., 2018)**

How large are these associations? **(Epskamp et al., 2018; Waldorp & Haslbeck, 2024)**

Which are supported as present or absent? **(Epskamp et al., 2018)**




Simulated data; regularized partial correlations.




Original estimates, bootstrap means, and central 95% bootstrap regions.
2,000 resamples; EBICglasso ($\gamma = .5$).

# Interpreting bootstrap results – a cofidence interval?

## What should a 95% confidence interval provide?

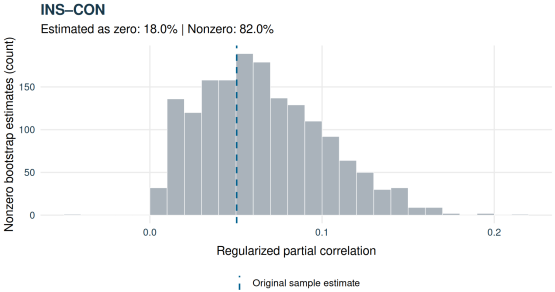
- Approximately 95% coverage of the population edge weight across repeated samples

## Why is regularization problematic? **(Waldorp & Haslbeck, 2024)**

- Bootstrap estimates inherit shrinkage and selection
- Central 95% regions need not achieve 95% coverage

## What does this limit?

- Bootstrap regions are not automatically valid confidence intervals or tests against zero
- The proportion of zero estimates is not the probability that the population edge is absent



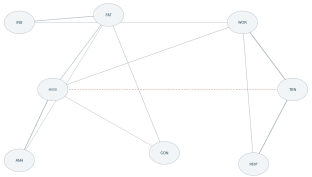

Histogram: nonzero bootstrap estimates only.
Dashed blue line: original sample estimate.

# Unregularized estimation with GGMnonreg (Williams, 2021)

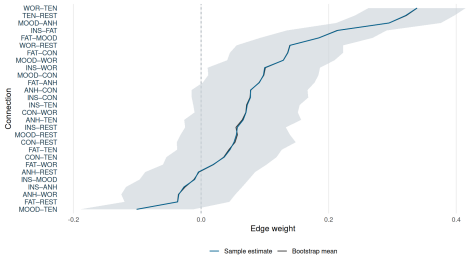
Which associations characterize the pattern?

How large are these associations?

Which are supported as present or absent?




Reduced network: edges retained when their pointwise 95% interval excludes zero.




All edge estimates and pointwise 95% bootstrap confidence intervals.

# Unregularized estimation with GGMnonreg (Williams, 2021)

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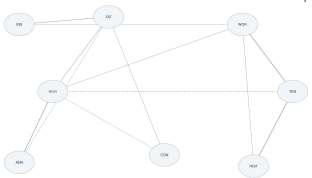
- Estimate all conditional associations without lasso shrinkage

## How large are these associations?

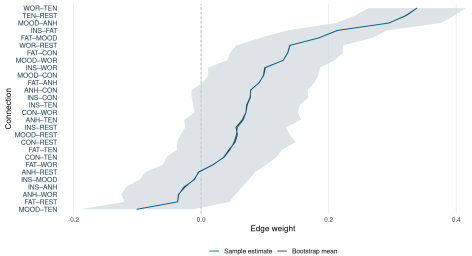
- Estimate magnitudes without shrinkage towards zero
- Approximate confidence intervals quantify estimation precision

## Which are supported as present or absent?

- Intervals excluding zero provide evidence against a zero association
- Intervals containing zero do not establish absence




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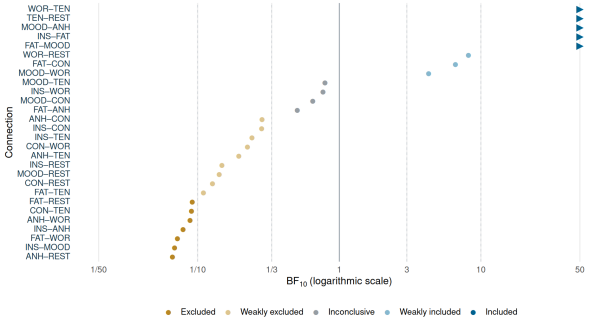

All edge estimates and pointwise 95% bootstrap confidence intervals.

# A Bayesian extension: Evidence for inclusion and exclusion (Williams & Mulder, 2020)

## What remains unresolved?

- An interval containing zero does not establish an absent association

## What can Bayesian analysis add?




Edgewise Bayes factors; evidence categories as in Huth et al. (2026). Triangles indicate values below 1/50 or above 50.

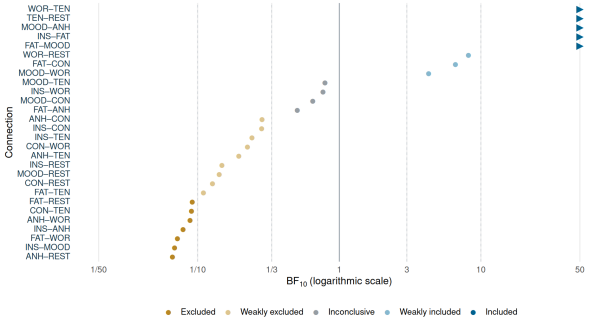
# A Bayesian extension: Evidence for inclusion and exclusion (Williams & Mulder, 2020)

## What remains unresolved?

- An interval containing zero does not establish an absent association

## What can Bayesian analysis add?

- Compare evidence for a nonzero association versus an absent edge
- Quantify support for either explanation, or identify inconclusive evidence
- so distinguish evidence favouring exclusion and insufficient evidence for conclusion




Edgewise Bayes factors; evidence categories as in Huth et al. (2026). Triangles indicate values below 1/50 or above 50.

## Research questions overview

| | Regularization + bootstrap | Unregularized + intervals | Bayesian estimation + Bayes factors |
|---|---|---|---|
| Characterize the association pattern | ✓ | ✓ | ✓ |
| Estimate association magnitudes and their uncertainty | △ Shrinkage; CI limitations | ✓ | ✓ |
| Evaluate evidence for presence **and** absence | – | △ Tests against zero only | ✓ |

✓ Can address the question under suitable assumptions  △ Partial answer / limitations  – Not provided by this workflow

**(Epskamp & Fried, 2018; Waldorp & Haslbeck, 2024; Williams, 2021; Williams & Mulder, 2020)**

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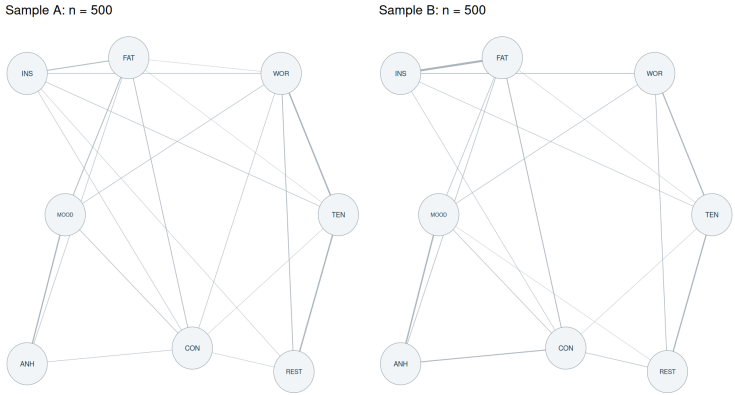
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**(Epskamp & Fried, 2018; Waldorp & Haslbeck, 2024; Williams, 2021; Williams & Mulder, 2020)**

- A method can support estimation without supporting every intended claim
- All approaches address estimation uncertainty, but differ in the conclusions they support

# Are these networks different beyond estimation variability?

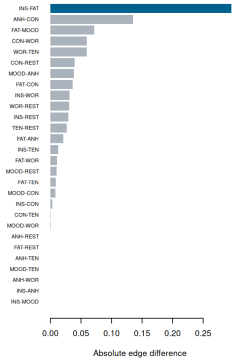



- Same eight symptoms; two independent samples of 500 participants
- Visible differences can reflect population differences, sampling variability, or both

# Three possible questions about stability in our symptom networks

## Network structure

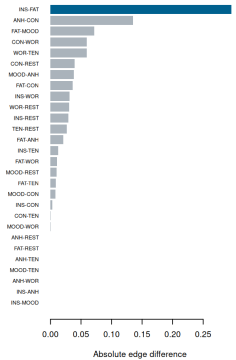
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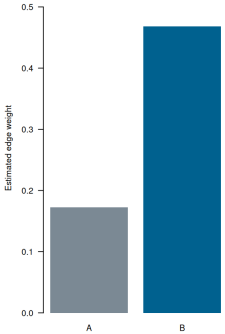
## Network structure

Does any symptom association differ?




## Specific edge

Does the INS–FAT association differ?

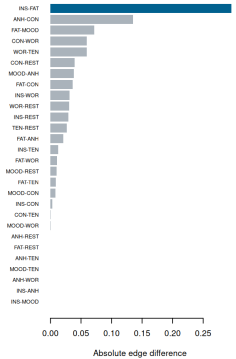


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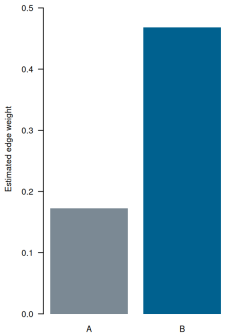
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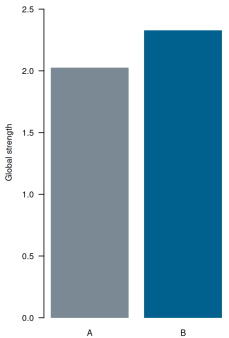

## Specific edge

Does the INS–FAT association differ?




## Global strength

Does overall connectivity differ?

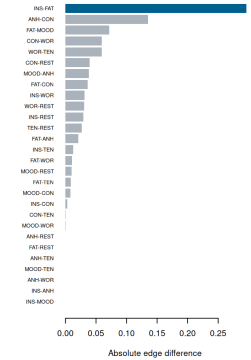


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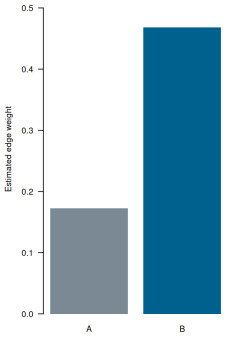
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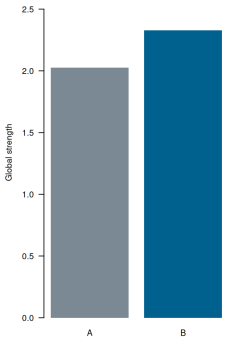

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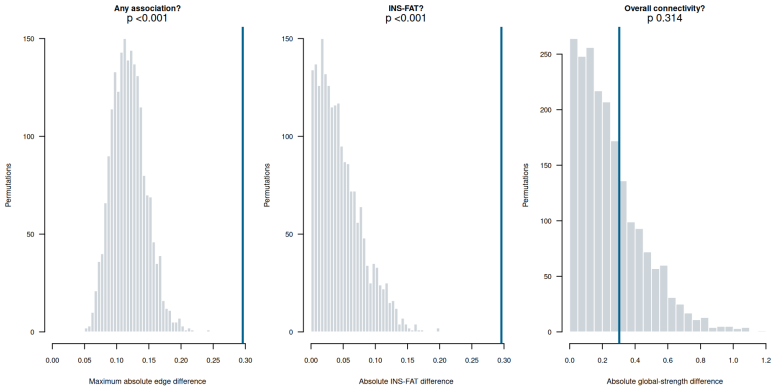



Are the observed differences larger than we would expect from sampling variation alone?

# Network Comparison Test: Are the observed differences unusual?

**(van Borkulo et al., 2023)**

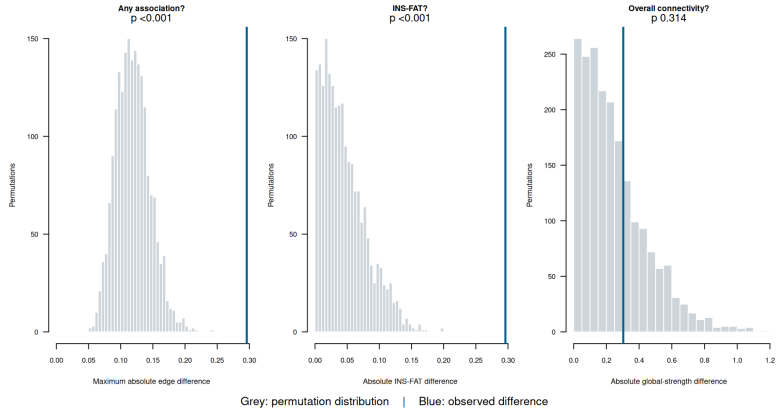
- Permute group labels → re-estimate both networks → recalculate all three comparison statistics.



# Network Comparison Test: Are the observed differences unusual?

**(van Borkulo et al., 2023)**

- Permute group labels → re-estimate both networks → recalculate all three comparison statistics.




- Observed difference in the upper tail → larger than typically expected from sampling variability under the null hypothesis

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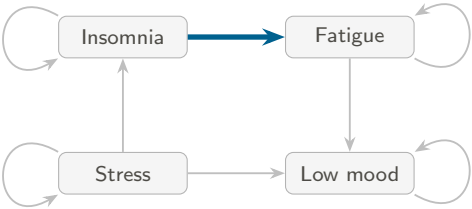
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# From cross-sectional to dynamic networks

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## Repeated measurements within persons (Epskamp, 2020)

- Temporal network: does insomnia at $t-1$ predict fatigue at $t$?
- Contemporaneous network: conditional associations within the same measurement occasion




Schematic temporal network.
Arrows: lagged effects ($t-1 \rightarrow t$); loops: autoregressive effects.

# From cross-sectional to dynamic networks

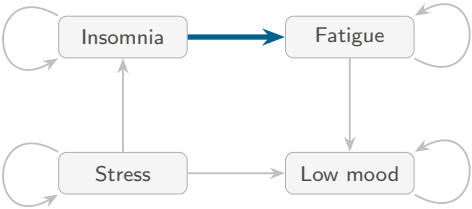
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## Many models, one representation

- VAR, graphicalVAR, mlVAR, DSEM (Asparouhov et al., 2018)
- Parameters of many dynamic models can be displayed as a network

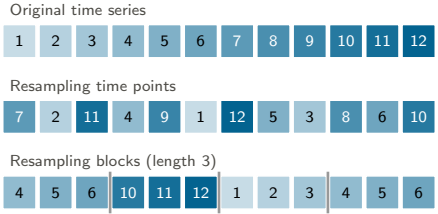



Schematic temporal network.
Arrows: lagged effects ($t-1 \rightarrow t$); loops: autoregressive effects.

# Estimation uncertainty: Why the familiar bootstrap breaks

## Resampling single time points

- Re-estimating on shuffled occasions removes the serial dependence the model is about




Numbers: measurement occasions.
Blocks keep short-term dependence intact.

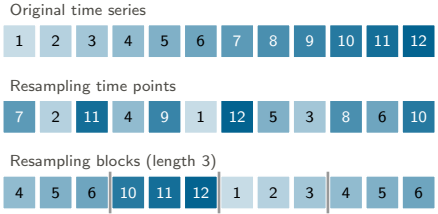
# Estimation uncertainty: Why the familiar bootstrap breaks

## Resampling single time points

- Re-estimating on shuffled occasions removes the serial dependence the model is about

## Regularized estimation (e.g., graphicalVAR)

- Same shrinkage and selection issues as in the cross-sectional case




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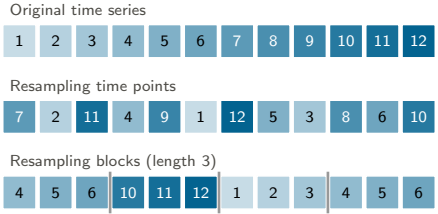
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## Limited uncertainty output

- Several implementations report no intervals for some or all parameters

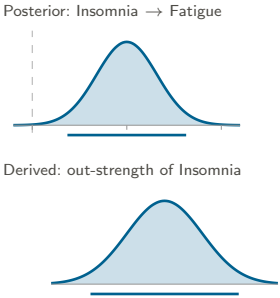



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# Quantifying uncertainty in dynamic networks

## Block bootstrap

- Resample contiguous blocks instead of single time points
- Block length must be chosen; gaps (e.g., nights) complicate block construction




Schematic posteriors with 95% credible intervals.
Same draws, any function of the parameters.

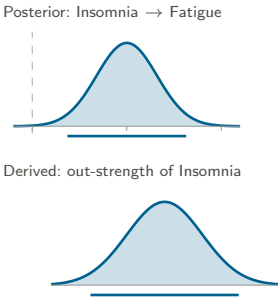
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- Posterior distribution for every parameter
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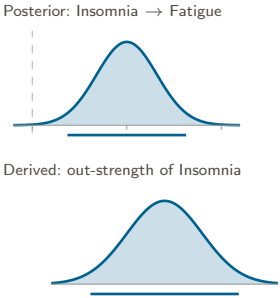
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## Carrying uncertainty forward (Siepe et al., 2026)

- Network features are often used in later analyses, e.g., predicting outcomes from centrality
- Taking their uncertainty into account leads to more realistic conclusions

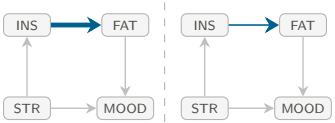



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# Different dynamic networks: beyond estimation variability?

**Two time series ($n = 1$)**

Is my insomnia → fatigue effect weaker after therapy?



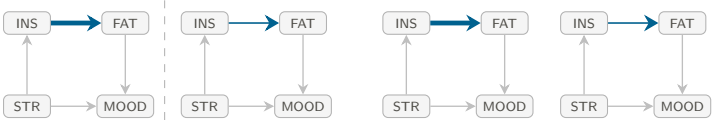
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**Two groups ($n > 1$)**

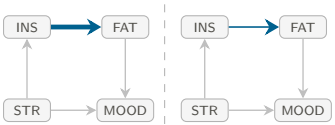
Does the average effect differ between therapy and control group?



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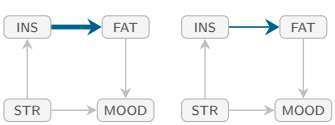
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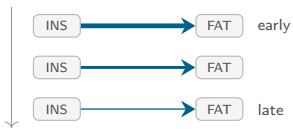

**Two groups ($n > 1$)**

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**Gradual change**

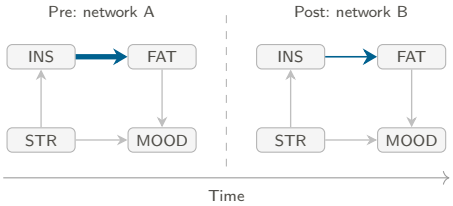
Does the effect change continuously over time?



# Comparing person-specific networks ($n = 1$)

## Typical questions

- Is my insomnia → fatigue effect weaker after therapy than before?
- Does my network differ from Kai's?




One person, two phases; each phase modeled separately.

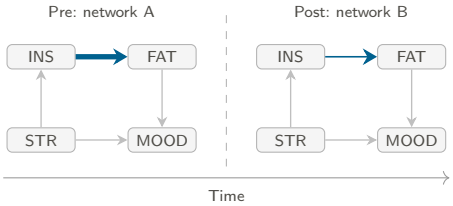
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## Data

- Two separate time series: two phases of one person, or two persons
- Each long enough for a person-specific model; often 100+ time points, depending on number of variables, effect sizes, and estimator **(Mansueto et al., 2023; Siepe et al., 2024; Zhang et al., 2025)**




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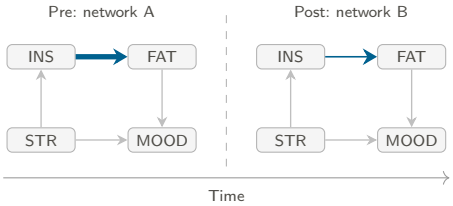
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## Assumption

- Parameters are constant within each series




One person, two phases; each phase modeled separately.

# Comparing person-specific networks: available tests

| Method | Approach | Answers |
|---|---|---|
| INIT (Hoekstra et al., 2024) | Compare model with equality constraints to unconstrained model (AIC/BIC) | Do the networks differ overall? |
| IVPP (Du et al., 2025) | Global test, then stepwise partial pruning of equality constraints | Which parameters differ? |
| Bayesian comparison (Siepe et al., 2024) | Posterior distance between networks vs. reference distribution of sampling variability | Does a network matrix differ beyond sampling variability? |
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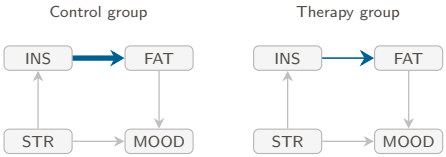
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- Global tests answer "does anything differ?", not "does insomnia → fatigue differ?"
- For a single effect: posterior (or block bootstrap distribution) of the difference

# Comparing groups of individuals ($n > 1$)

## Typical questions

- Does the average insomnia → fatigue effect differ between therapy and control group?
- Do individuals differ more from each other in one group?




Average (fixed-effect) temporal networks; individual networks vary around them.

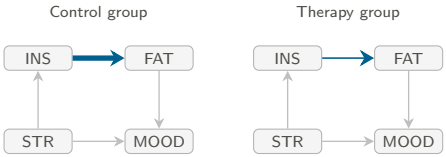
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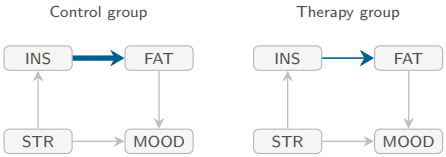
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## What can differ?

- Fixed effects, random-effect variances, contemporaneous and between-person networks

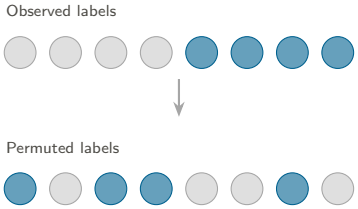



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# Comparing groups: available approaches

## Permutation test for mlVAR (Haslbeck et al., 2025)

- Permute group labels, re-estimate, recompute differences
- The dynamic counterpart of the NCT




Re-estimate both groups;
repeat many times → reference distribution

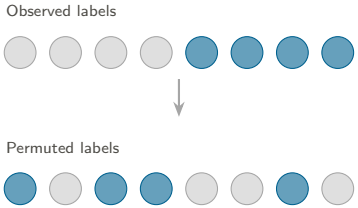
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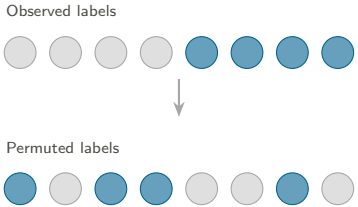
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### Multilevel models with group predictors (Asparouhov et al., 2018)

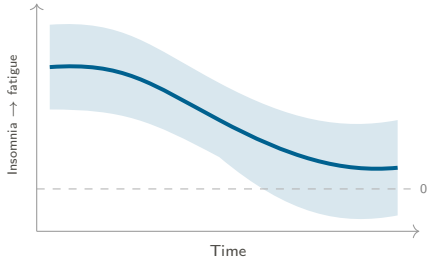
- DSEM or Bayesian mlVAR: group membership predicts person-specific parameters
- Posterior of the group difference for any parameter or derived quantity



# Gradual change: time-varying networks

## Typical questions

- Does the insomnia → fatigue effect weaken gradually during therapy?
- Does insomnia become less central over time?




Schematic time-varying effect with pointwise uncertainty band.

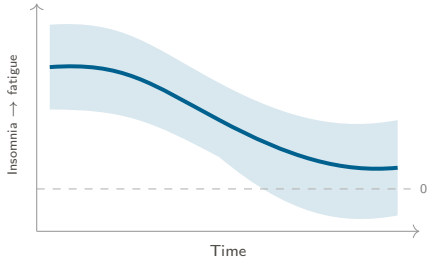
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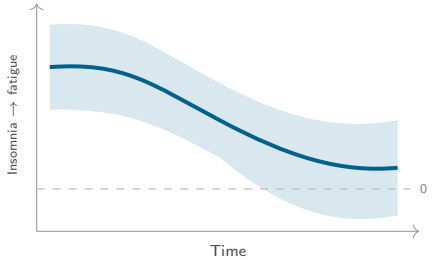
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- Time-varying VAR: let parameters change smoothly
- Moderation: parameters as a function of a measured variable (e.g., treatment phase)

## Caveats

- Requires long time series; flexibility trades off against precision
- Moderation requires specific hypothesesl




Schematic time-varying effect with pointwise uncertainty band.

# Outline

1 Terminology and Motivation

2 Cross-sectional Networks

3 Longitudinal networks

4 Conclusion

# Take-home messages

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## Stability of estimation

- Quantify uncertainty for the quantity you interpret: an edge, a centrality index, a difference
- Methods support different claims: bootstrap regions, confidence intervals, and Bayes factors are not interchangeable
- Time series need dependence-aware methods, e.g., block bootstrap or Bayesian estimation

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Start from the research question: which part of the network should be (in)stable, and why does it matter?

## DYNAMIC Methods Consultation

### Find resources

- **DYNAMIC website**
  https://www.dynamic-center.net/forschung/methodenberatung/
- **FAQ page**
  https://pandar.netlify.app/extras/main/#dynamic

### Stay informed

- Subscribe to our methods newsletter ( bjoern.siepe@uni-marburg.de)

### Get personal support

- Contact us via our Google Form
- Online consultation meetings
- Written feedback on your questions

**Slides**

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